NAME: PATEL GRENA M.

ROLL NO:3239

DIV: C

JAVA ASSIGNMENT 2

Q1

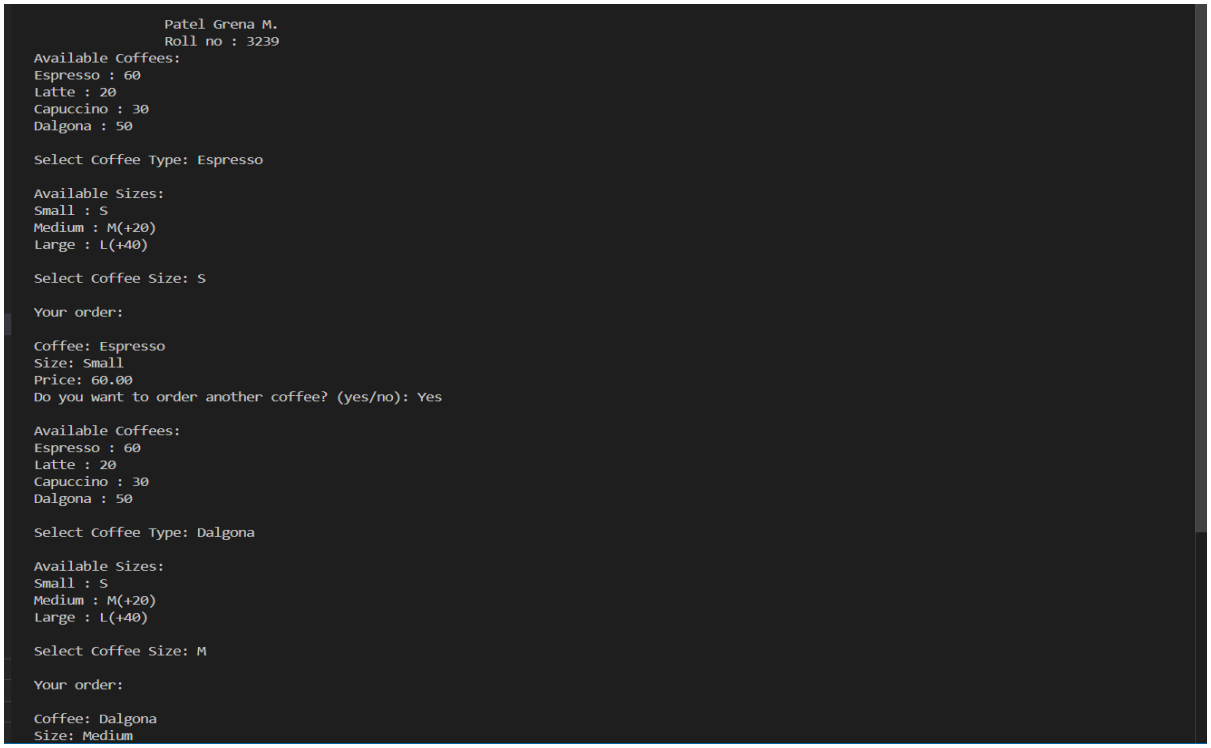
```
package com.cafe;
import java.util.Scanner;
import java.util.ArrayList;
public interface coffeeOrder{
  public double getPrice();
  public void selectCoffee(String Type);
  public void selectSize(String size);
  public void displayOrder();
}
class cafeOrder implements coffeeOrder{
    String coffeeType;
    String coffeeSize;
    double price;
    public void selectCoffee(String Type){
        Type=Type.toLowerCase();
    if(Type.equals("dalgona")){
        price=50;
        coffeeType="Dalgona";
    }
    else if(Type.equals("latte")){
        price=20;
        coffeeType="Latte";
    }
    else if(Type.equals("capuccino")){
        price=30;
```

```
    coffeeType="Capuccino";
  }
  else if(Type.equals("espresso")){
    price=60;
    coffeeType="Espresso";
  }
  }
  public void selectSize(String size){
    size=size.toLowerCase();
    if(size.equals("s")){
      coffeeSize="Small";
    }
    else if(size.equals("m")){
      price+=20;
      coffeeSize="Medium";
    }
    else if(size.equals("l")){
      price+=40;
      coffeeSize="Large";
    }
    else{
      System.out.println("Please select correct size.");
    }
  }
  public void availableCoffee() {
    System.out.println("\nAvailable Coffees:");
    System.out.println("Espresso : 60");
    System.out.println("Latte : 20");
    System.out.println("Cappuccino : 30");
    System.out.println("Dalgona : 50");
 }
```

```
    public void availableSize() {
        System.out.println("\nAvailable Sizes:");
        System.out.println("Small : S");
        System.out.println("Medium : M(+20)");
        System.out.println("Large : L(+40)");
    }
    public double getPrice() {
        return price;
    }
    public void displayOrder() {
        System.out.printf("\nCoffee: %s\nSize: %s\nPrice: %.2f",coffeeType,coffeeSize,price);
    }
}
class Main{
    public static void main(String args[]){
      Scanner s=new Scanner(System.in);
      String choice;
      double total=0;
      ArrayList<cafeOrder> orders=new ArrayList<>();
      System.out.print("\n\t\t\Patel Grena M.");
      System.out.print("\n\t\tRoll no : 3239");
      do{
          cafeOrder c=new cafeOrder();
          c.availableCoffee();
          System.out.print("\nSelect Coffee Type: ");
          String type=s.nextLine();
          c.selectCoffee(type);
          c.availableSize();
          System.out.print("\nSelect Coffee Size: ");
          String size = s.nextLine();
          c.selectSize(size);
```

```
            System.out.println("\nYour order:");
            c.displayOrder();
            orders.add(c);
            System.out.print("\nDo you want to order another coffee? (yes/no): ");
            choice = s.nextLine();
        }while(choice.equalsIgnoreCase("yes"));
        System.out.println("\nTotal Orders information...:");
        for (int i = 0; i < orders.size(); i++) {
            System.out.print("\n\nOrder " + (i + 1) + ": ");
            orders.get(i).displayOrder();
            total += orders.get(i).getPrice();
        }
        System.out.printf("\nTotal Price: %.2f\n", total);
        s.close();
    }
}
```

OUTPUT:



```
Select Coffee Size: S

Your order:

Coffee: Espresso
Size: Small
Price: 60.00
Do you want to order another coffee? (yes/no): Yes

Available Coffees:
Espresso : 60
Latte : 20
Capuccino : 30
Dalgona : 50

Select Coffee Type: Dalgona

Available Sizes:
Small : S
Medium : M(+20)
Large : L(+40)

Select Coffee Size: M

Your order:

Coffee: Dalgona
Size: Medium
Price: 70.00
Do you want to order another coffee? (yes/no): no

Total Orders information...:


Order 1:
Coffee: Espresso
Size: Small
Price: 60.00

Order 2:
Coffee: Dalgona
Size: Medium
Price: 70.00
Total Price: 130.00
```

Q2

```
package com.kiosk;

import java.util.Scanner;


public interface DenominationHandler {
  public void nexthandler(DenominationHandler dm);
  public void handle(int amt);
}


public class FiveHundredHandler implements DenominationHandler{
   DenominationHandler dm;
   public void nexthandler(DenominationHandler dm) {
     this.dm=dm;
   }
```

```
    public void handle(int amt) {
      if(amt>=500){
       int cn=amt/500;
       System.out.println("\t\t500     "+cn);
       amt%=500;
      }
      if(amt!=0){
       dm.handle(amt);
      }
    }

}
public class OneHundredHandler implements DenominationHandler{
    DenominationHandler dm;
    public void nexthandler(DenominationHandler dm) {
      this.dm=dm;
    }
    public void handle(int amt) {
      if(amt>=100){
       int cn=amt/100;
       System.out.println("\t\t100     "+cn);
       amt%=100;
      }
      if(amt!=0){
       dm.handle(amt);
      }
    }

}

public class TenHandler implements DenominationHandler{
```

```
    DenominationHandler dm;
    public void nexthandler(DenominationHandler dm) {
      this.dm=dm;
    }
    public void handle(int amt) {
      if(amt>=10){
       int cn=amt/10;
       System.out.println("\t\t10      "+cn);
       amt%=10;
      }
      if(amt!=0){
       dm.handle(amt);
      }
    }

}
public class FiveHandler implements DenominationHandler{
    DenominationHandler dm;
    public void nexthandler(DenominationHandler dm) {
      this.dm=dm;
    }
    public void handle(int amt) {
      if(amt>=5){
       int cn=amt/5;
       System.out.println("\t\t5       "+cn);
       amt%=5;
      }
      if(amt<5){
       System.out.println("\nAmount(Remaining) : "+amt);
      }
    }
```

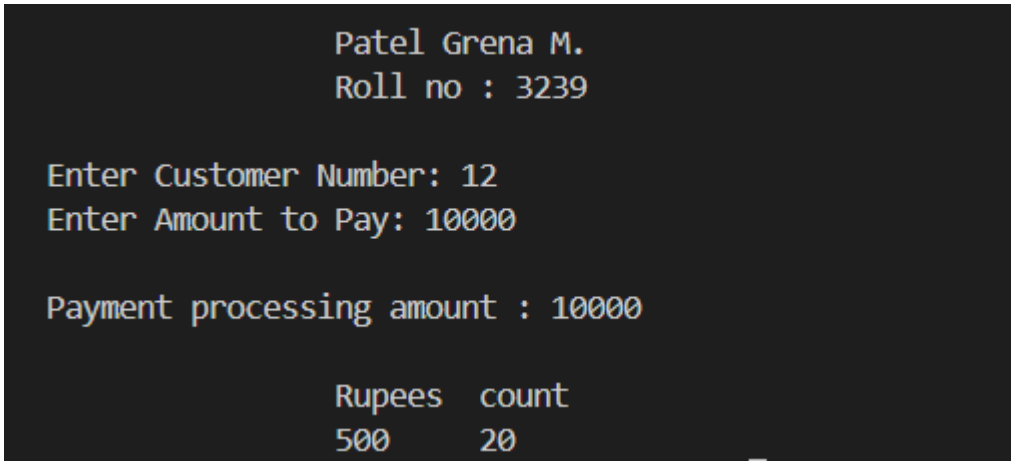
```
}
public class Kiosk {
   final DenominationHandler chain;
   Kiosk(){
      FiveHundredHandler fh=new FiveHundredHandler();
      OneHundredHandler oh=new OneHundredHandler();
      TenHandler t=new TenHandler();
      FiveHandler f=new FiveHandler();
      fh.nexthandler(oh);
      oh.nexthandler(t);
      t.nexthandler(f);
      this.chain=fh;
   }
   void pay(int amt){
      System.out.println("\nPayment processing amount : "+amt);
      System.out.print("\n\t\tRupees  count\n");
      chain.handle(amt);
   }
}

public class Main {
   public static void main(String[] args) {
      Scanner s = new Scanner(System.in);
      System.out.print("\n\t\TPatel Grena M.");
      System.out.println("\n\t\tRoll no : 3239\n");
      System.out.print("Enter Customer Number: ");
      String cNo = s.nextLine();

      System.out.print("Enter Amount to Pay: ");
      int amt = s.nextInt();
```

```
        Kiosk k =new Kiosk();
        k.pay(amt);
        s.close();
    }
}
```

OUTPUT:



Q3

```
package com.bank;
package com.bank.exceptions;
import java.util.Scanner;
import com.bank.exceptions.BankingException;
public class BankingException extends Exception{
    public BankingException(String msg){
        super(msg);
    }
}
public interface BankingOperations {
    public void deposit(double amt) throws BankingException;
    public void withdraw(double amt) throws BankingException;
    double getBalance();
}
```

```
public class Account implements BankingOperations {
    String accNo;
    double balance = 0.0;

    Account(String accNo) {
        this.accNo = accNo;
    }

    public void deposit(double amt) throws BankingException {
        if (amt <= 0) {
            throw new BankingException("Deposit amount must be greater than zero.");
        }
        balance+=amt;
        System.out.println("Successfully deposited Rs." + amt);
    }

    public void withdraw(double amt) throws BankingException {
       if(amt<=0){
        throw new BankingException("Amount of withdraw must be greater than zero.");
       }
        if (amt > balance) {
            throw new BankingException("No sufficient amount to withdraw");
        }
        balance-=amt;
        System.out.println("Successfully withdraw Rs." + amt);
    }

    public double getBalance() {
         return balance;
    }
```

```
    public String getAccNo() {
        return accNo;
    }
}

public class BankingSystem {
    final int Max_acc=10;
    private Account[] ac;
    private int count;
    BankingSystem(){
        ac=new Account[Max_acc];
        count=0;
    }
    Account findAcc(String accNo){
        for(int i=0;i<count;i++){
            if(ac[i].getAccNo().equals(accNo)){
                return ac[i];
            }
        }
        return null;
    }
    void createAcc(String accNo) throws BankingException{
        if(count>=Max_acc){
            throw new BankingException("Reached Account limit already.");
        }
        if(findAcc(accNo)!=null){
            throw new BankingException("Account already exists.");
        }
        ac[count++]=new Account(accNo);
        System.out.println("Account created successfully with Account Number: " + accNo);
    }
```

```
    public Account getAcc(String accNo) throws BankingException {
        Account acc = findAcc(accNo);
        if (acc == null) {
            throw new BankingException("Account not found.");
        }
        return acc;
    }
}

public class Main {
    public static void main(String[] args) {
        BankingSystem bs = new BankingSystem();
        Scanner s = new Scanner(System.in);
        System.out.print("\n\t\tPatel Grena M.");
        System.out.println("\n\t\tRoll no : 3239");
        int ch = 0;
        while (ch <= 5) {
            System.out.println("\n_______Banking System_______\n");
            System.out.println("1. Create Account");
            System.out.println("2. Deposit Money");
            System.out.println("3. Withdraw Money");
            System.out.println("4. Check Balance");
            System.out.println("5. Exit");
            System.out.print("Enter your choice: ");
            ch = s.nextInt();
            s.nextLine();
            try {
                switch (ch) {
                    case 1:
                        System.out.print("Enter Account Number: ");
                        String accNo = s.nextLine();
```

```
            bs.createAcc(accNo);
            break;
        case 2:
            System.out.print("Enter Account Number: ");
            accNo = s.nextLine();
            Account account = bs.getAcc(accNo);
            System.out.print("Enter Deposit Amount: Rs.");
            double depAmt = s.nextDouble();
            account.deposit(depAmt);
            break;
        case 3:
            System.out.print("Enter Account Number: ");
            accNo = s.nextLine();
            account = bs.getAcc(accNo);
            System.out.print("Enter Withdrawal Amount: Rs.");
            double withDrawAmt = s.nextDouble();
            account.withdraw(withDrawAmt);
            break;
        case 4:
            System.out.print("Enter Account Number: ");
            accNo = s.nextLine();
            account = bs.getAcc(accNo);
            System.out.println("Current Balance: Rs." + account.getBalance());
            break;
        case 5:
            System.out.println("Thank you for using the Banking System!");
            return;
        default:
            System.out.println("Invalid choice. Please try again.");
    }
} catch (BankingException e) {
```

```
            System.out.println("Error: " + e.getMessage());
        }
    }
    s.close();
  }
}
```

OUTPUT:

```
              Patel Grena M.
              Roll no : 3239

______Banking System______

1. Create Account
2. Deposit Money
3. Withdraw Money
4. Check Balance
5. Exit
Enter your choice: 1
Enter Account Number: 12
Account created successfully with Account Number: 12

______Banking System______

1. Create Account
2. Deposit Money
3. Withdraw Money
4. Check Balance
5. Exit
Enter your choice: 2
Enter Account Number: 12
Enter Deposit Amount: Rs.5000
Successfully deposited Rs.5000.0

______Banking System______

1. Create Account
2. Deposit Money
3. Withdraw Money
4. Check Balance
5. Exit
Enter your choice: 4
Enter Account Number: 12
Current Balance: Rs.5000.0

______Banking System______

1. Create Account
2. Deposit Money
3. Withdraw Money
4. Check Balance
```

```
Successfully deposited Rs.5000.0

_______Banking System_______

1. Create Account
2. Deposit Money
3. Withdraw Money
4. Check Balance
5. Exit
Enter your choice: 4
Enter Account Number: 12
Current Balance: Rs.5000.0

_______Banking System_______

1. Create Account
2. Deposit Money
3. Withdraw Money
4. Check Balance
5. Exit
Enter your choice: 3
Enter Account Number: 12
Enter Withdrawal Amount: Rs.2000
Successfully withdraw Rs.2000.0

_______Banking System_______

1. Create Account
2. Deposit Money
3. Withdraw Money
4. Check Balance
5. Exit
Enter your choice: 4
Enter Account Number: 12
Current Balance: Rs.3000.0

_______Banking System_______

1. Create Account
2. Deposit Money
3. Withdraw Money
4. Check Balance
5. Exit
Enter your choice: 5
Thank you for using the Banking System!
```

Q4

```
package hospital;

import java.util.Scanner;


public abstract class Doctor {
    private String name;
    private String specialization;
    public Doctor(String name, String specialization) {
        this.name = name;
        this.specialization = specialization;
    }
    public String getName() {
        return name;
    }
```

```
    public String getSpec() {
        return specialization;
    }
    public abstract void bookAppointment();
}

public class Specialist extends Doctor {
    private String expertise;
    public Specialist(String name, String sp, String expertise) {
        super(name, sp);
        this.expertise = expertise;
    }
    public String getExpertise() {
        return expertise;
    }
    public void bookAppointment() {
    System.out.println("\n\tBooking an appointment...\n");
    System.out.println("\tDoctor Name: " + getName());
    System.out.println("\tSpecialized Field: " + getSpec());
    System.out.println("\tExpertise: " + getExpertise()+"\n");
    }
}

public class Appointment {
        private Doctor dr;

        public Appointment(Doctor dr) {
            this.dr = dr;
        }

        public void schedule() {
```

```
        dr.bookAppointment();
    }
}

public class Main {
    public static void main(String[] args) {
        Scanner s= new Scanner(System.in);
        System.out.print("\n\t\tPatel Grena M.");
        System.out.println("\n\t\tRoll no : 3239\n");
        System.out.print("Enter doctor's name:");
        String name = s.nextLine();
        System.out.print("Enter doctor's specialization:");
        String specialization = s.nextLine();

        System.out.print("Enter doctor's expertise:");
        String expertise = s.nextLine();

        Specialist specialist = new Specialist(name, specialization, expertise);
        Appointment at = new Appointment(specialist);
        at.schedule();
        s.close();
   }
}
```

OUTPUT:

```
                Patel Grena M.
                Roll no : 3239

Enter doctor's name:Rajen Shah
Enter doctor's specialization:ENT specialist
Enter doctor's expertise:Tonsil opd

        Booking an appointment...

        Doctor Name: Rajen Shah
        Specialized Field: ENT specialist
        Expertise: Tonsil opd
```

Q5

```
import java.util.ArrayList;
import java.util.Iterator;
import java.util.List;
import java.util.Scanner;

class BankAccount {
  private String accountHolder;
  private double balance = 0.0;
  private List<String> transactions;

  BankAccount(String var1) {
    this.accountHolder = var1;
    this.transactions = new ArrayList();
    this.transactions.add("Initial account balance: Rs.0.0");
  }

  void deposit(double var1) {
    if (var1 > 0.0) {
      this.balance += var1;
      this.transactions.add("Deposited : Rs." + var1);
      System.out.println("Successfully deposited: Rs." + var1);
    } else {
      System.out.println("Deposit amount must be positive.");
    }

  }

  public void withdraw(double var1) {
    if (var1 > 0.0) {
      if (var1 <= this.balance) {
```

```
            this.balance -= var1;
            this.transactions.add("Withdrawn: Rs." + var1);
            System.out.println("Successfully Withdraw amount : Rs." + var1);
        } else {
            System.out.println("Insufficient funds. Current balance: Rs." + this.balance);
        }
    } else {
        System.out.println("Withdrawal amount must be positive.");
    }

  }

  public double getBalance() {
    return this.balance;
  }

  public void printMiniStatement() {
    System.out.println("\n--- Mini Statement ---");
    Iterator var1 = this.transactions.iterator();

    while(var1.hasNext()) {
      String var2 = (String)var1.next();
      System.out.println(var2);
    }

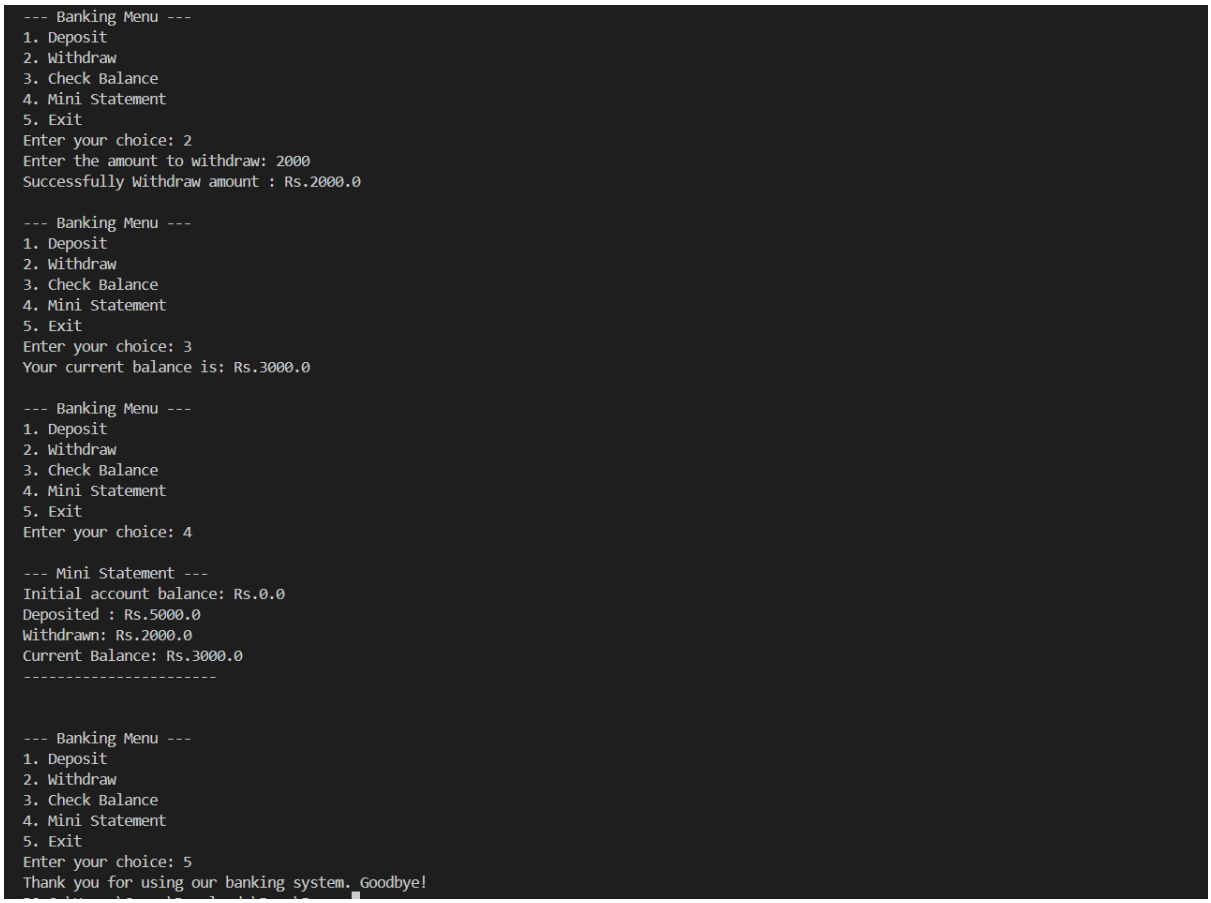
    System.out.println("Current Balance: Rs." + this.balance);
    System.out.println("----------------------\n");
  }
}
class BankAccount {
   private String accountHolder;
```

```
private double balance=0.0;
private List<String> transactions;
BankAccount(String nm){
   this.accountHolder=nm;
   this.transactions = new ArrayList<>();
   transactions.add("Initial account balance: Rs.0.0");
}
void deposit(double amt){
 if(amt>0){
   balance+=amt;
   transactions.add("Deposited : Rs."+amt);
   System.out.println("Successfully deposited: Rs." + amt);
   }
 else {
       System.out.println("Deposit amount must be positive.");
 }
}
public void withdraw(double amt) {
   if (amt > 0) {
      if (amt <= balance) {
         balance -= amt;
         transactions.add("Withdrawn: Rs." + amt);
         System.out.println("Successfully Withdraw amount : Rs." + amt);
      }
       else {
         System.out.println("Insufficient funds. Current balance: Rs." + balance);
      }
   } else {
      System.out.println("Withdrawal amount must be positive.");
   }
}
```

```
    public double getBalance() {
        return balance;
    }
    public void printMiniStatement() {
        System.out.println("\n--- Mini Statement ---");
        for (String tr : transactions) {
            System.out.println(tr);
        }
        System.out.println("Current Balance: Rs." + balance);
        System.out.println("-----------------------\n");
    }
}
public class BankingSystem {
    public static void main(String[] args) {
        Scanner s = new Scanner(System.in);
        System.out.print("\n\t\tPatel Grena M.");
        System.out.println("\n\t\tRoll no : 3239\n");
        System.out.print("Enter your name : ");
        String name = s.nextLine();
        BankAccount account = new BankAccount(name);
        System.out.println("Welcome, " + name + "! Your bank account has been created.");
        int choice;
        do {
            System.out.println("\n--- Banking Menu ---");
            System.out.println("1. Deposit");
            System.out.println("2. Withdraw");
            System.out.println("3. Check Balance");
            System.out.println("4. Mini Statement");
            System.out.println("5. Exit");
            System.out.print("Enter your choice: ");
            while (!s.hasNextInt()) {
```

```
            System.out.print("Invalid input. Enter a valid choice: ");
            s.next();
        }
        choice = s.nextInt();
        switch (choice) {
            case 1:
                System.out.print("Enter the amount to deposit: ");
                double depositAmount = s.nextDouble();
                account.deposit(depositAmount);
                break;
            case 2:
                System.out.print("Enter the amount to withdraw: ");
                double withdrawAmount = s.nextDouble();
                account.withdraw(withdrawAmount);
                break;
            case 3:
                System.out.println("Your current balance is: Rs." + account.getBalance());
                break;
            case 4:
                account.printMiniStatement();
                break;
            case 5:
                System.out.println("Thank you for using our banking system. Goodbye!");
                break;
            default:
                System.out.println("Invalid choice. Please try again.");
        }
    } while (choice != 5);
    s.close();
  }
}
```

OUTPUT:



Q6

```
package propertymanagement;

import java.util.ArrayList;


public class Apartments implements Property{
   String location;
   double price;
   public Apartments(String loc, double pr) {
      this.location = loc;
      this.price = pr;
      }
   public void buy() {
     System.out.println("Buying an Apartment in "+location+" of Rs."+price);
   }
```

```
   public void sell() {
    System.out.println("Selling an Apartment in "+location+" of Rs."+price);
    }
}

public interface Property {
    void buy();
    void sell();
}

public class Bungalow implements Property{
    String location;
    double price;
    public Bungalow(String loc, double pr) {
        this.location = loc;
        this.price = pr;
        }
    public void buy() {
       System.out.println("Buying a Bunglow in "+location+" of Rs."+price);
    }
   public void sell() {
    System.out.println("Selling a Bunglow in "+location+" of Rs."+price);
    }
}

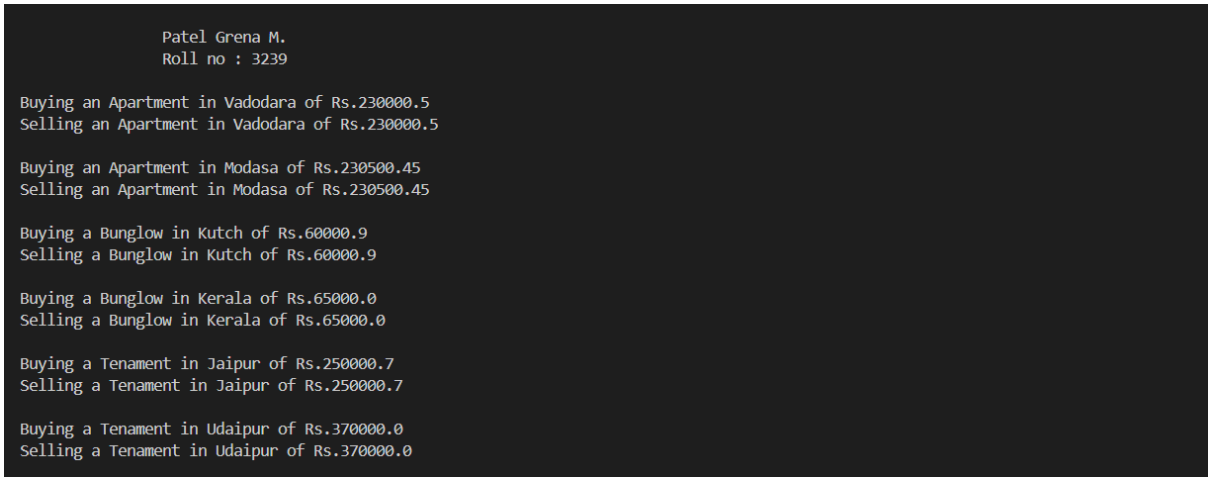
public class Tenaments implements Property{
    String location;
    double price;
    public Tenaments(String loc, double pr) {
        this.location = loc;
        this.price = pr;
```

```
    }
  public void buy() {
    System.out.println("Buying a Tenament in "+location+" of Rs."+price);
  }
 public void sell() {
  System.out.println("Selling a Tenament in "+location+" of Rs."+price);
  }
}

public class Application {
   public static void main(String[] args) {
     ArrayList<Property> pr = new ArrayList<>();
     System.out.print("\n\t\tPatel Grena M.");
     System.out.println("\n\t\tRoll no : 3239\n");
     Apartments a=new Apartments("Vadodara", 230000.50);
     Apartments ap=new Apartments("Modasa", 230500.45);
     Bungalow b=new Bungalow("Kutch", 60000.90);
     Bungalow bg=new Bungalow("Kerala", 65000.00);
     Tenaments t=new Tenaments("Jaipur", 250000.70);
     Tenaments tn=new Tenaments("Udaipur", 370000.00);
     pr.add(a);
     pr.add(ap);
     pr.add(b);
     pr.add(bg);
     pr.add(t);
     pr.add(tn);
     for(Property p:pr){
      p.buy();
      p.sell();
      System.out.println();
     }
```

```
  }
}
```

OUTPUT:



Q7

```
public class Flour {
   double weight;
   double price;
   Flour(double w,double p){
      this.weight=w;
      this.price=p;
   }
   double getW(){
      return this.weight;
   }
   double getP(){
      return this.price;
   }
   double cal(){
      return price*weight;
   }
}
```

```
public class FlourItem {

    Flour getF(){
        return new Flour(1, 45.50);
    }

}

public interface FlourItemInterface {

    Flour getQuintal();
    Flour get10kg();
    Flour get1kg();

}

public class FlourStore implements FlourItemInterface {
    public Flour getQuintal() {
        return new Flour(100, 90.10);
    }
    public Flour get10kg() {
        return new Flour(10, 105.0);
    }
    public Flour get1kg() {
        return new Flour(1, 110.0);
    }
}

public class Application {
    public static void main(String[] args) {
        FlourStore s = new FlourStore();
```

```
        Flour q = s.getQuintal();
        Flour t = s.get10kg();
        Flour o = s.get1kg();
        System.out.print("\n\t\tPatel Grena M.");
        System.out.println("\n\t\tRoll no : 3239\n");

        System.out.println("100 kg Flour Packet:");
        System.out.println("Weight: " + q.getW() + " kg");
        System.out.println("Total Price: " + q.cal() + " Rs.\n");

        System.out.println("10 kg Flour Packet:");
        System.out.println("Weight: " + t.getW() + " kg");
        System.out.println("Total Price: " + t.cal() + " Rs.\n");

        System.out.println("1 kg Flour Packet:");
        System.out.println("Weight: " + o.getW() + " kg");
        System.out.println("Total Price: " + o.cal() + " Rs.");
    }
}
```

OUTPUT:

```
                Patel Grena M.
                Roll no : 3239

100 kg Flour Packet:
Weight: 100.0 kg
Total Price: 9010.0 Rs.

10 kg Flour Packet:
Weight: 10.0 kg
Total Price: 1050.0 Rs.

1 kg Flour Packet:
Weight: 1.0 kg
Total Price: 110.0 Rs.
```

Q8

```
package library;
import java.util.Scanner;

public abstract class AbstractBook {
  private String title;
  private String author;
  private boolean isLent;
   AbstractBook(String t,String w){
      this.title=t;
      this.author=w;
      this.isLent=false;
   }
   String getT(){
      return title;
   }
   String getA(){
      return author;
   }
   boolean isLent(){
      return isLent;
   }
   void lend(){
   if(!isLent){
      isLent=true;
      System.out.println(getT()+ " has been lent.");
      }
   else {
      System.out.println(getT()+ " is already lent out.");
   }
   }
```

```
    void ret(){
        if(isLent){
            isLent=true;
            System.out.println(getT()+ " has been returned.");
            }
        else {
            System.out.println(getT()+ " was not lent out.");
        }
    }
    abstract String getBookType();
     public String toString() {
    return "Title: " + title + "\n  Author: " + author + "\n  Is Lent: " + (isLent ? "Yes" : "No");
}

}

public class Book extends AbstractBook {
    Book(String t,String w){
       super(t, w);
    }
   String getBookType(){
    return "Regular Book";
    }
}

public class Library {
    AbstractBook[] book;
    int cn;
    Library(){
        book=new AbstractBook[100];
        cn=0;
```

```
}
void addBook(AbstractBook b){
if(cn<book.length){
   book[cn++]=b;
   System.out.println("Book added: " + b.getT());
}
else{
   System.out.println("Library is full, cannot add more books.");
}
}
void lendBook(String t){
  for(int i=0;i<cn;i++){
   if(book[i].getT().equalsIgnoreCase(t)){
    book[i].lend();
    return;
   }
   System.out.println(t + " is not available in the library.");
  }
}
void returnBook(String t){
   for(int i=0;i<cn;i++){
       if(book[i].getT().equalsIgnoreCase(t)){
        book[i].ret();
        return;
       }
       System.out.println(t + " is not found in the library.");
      }
}
void listBooks(){
   if (cn==0) {
       System.out.println("No books in the library.");
```

```
            return;
        }
        else{
            System.out.println("-----Book List-----");
         for (int i = 0; i < cn; i++) {
            System.out.println((i+1)+"."+book[i]);
         }
        }
    }
}

public class Application {
   public static void main(String[] args) {
        Library library = new Library();
        Scanner s = new Scanner(System.in);
        int ch;
        System.out.print("\n\t\tPatel Grena M.");
        System.out.print("\n\t\tRoll no : 3239\n");
        do {
            System.out.println("\nLibrary Management System");
            System.out.println("1. Add Book");
            System.out.println("2. Lend Book");
            System.out.println("3. Return Book");
            System.out.println("4. List All Books");
            System.out.println("5. Exit");
            System.out.print("Choose an option: ");
            ch = s.nextInt();
            s.nextLine();

            switch (ch) {
                case 1:
```

```
                System.out.print("Enter book title: ");
                String t = s.nextLine();
                System.out.print("Enter book author: ");
                String a = s.nextLine();
                library.addBook(new Book(t, a));
                break;
            case 2:
                System.out.print("Enter title of the book to lend: ");
                String lt = s.nextLine();
                library.lendBook(lt);
                break;
            case 3:
                System.out.print("Enter title of the book to return: ");
                String ret = s.nextLine();
                library.returnBook(ret);
                break;
            case 4:
                library.listBooks();
                break;
            case 5:
                System.out.println("Exiting....");
                break;
            default:
                System.out.println("Invalid choice. Please try again.");
        }
    } while (ch != 5);
    s.close();
  }
}
```

OUTPUT:

```
                    Patel Grena M.
                    Roll no : 3239

   Library Management System
   1. Add Book
   2. Lend Book
   3. Return Book
   4. List All Books
   5. Exit
   Choose an option: 1
   Enter book title: HARRY POTER
   Enter book author: DIVYA JYOTI
   Book added: HARRY POTER

   Library Management System
   1. Add Book
   2. Lend Book
   3. Return Book
   4. List All Books
   5. Exit
   Choose an option: 2
   Enter title of the book to lend: HARRY POTER
   HARRY POTER has been lent.

   Library Management System
   1. Add Book
   2. Lend Book
   3. Return Book
   4. List All Books
   5. Exit
   Choose an option: 4
   -----Book List-----
```

```
Choose an option: 4
-----Book List-----
1.Title: HARRY POTER
  Author: DIVYA JYOTI
  Is Lent: Yes

Library Management System
1. Add Book
2. Lend Book
3. Return Book
4. List All Books
5. Exit
Choose an option: 3
Enter title of the book to return: HARRY POTER
HARRY POTER has been returned.

Library Management System
1. Add Book
2. Lend Book
3. Return Book
4. List All Books
5. Exit
Choose an option: 5
Exiting....
```

Q9

```
<html>
   <body>
      <applet code="Q9.class"height="100" width="100"></applet>
   </body>
</html>
import java.applet.Applet;
import java.awt.*;
import java.awt.event.*;

public class q9 extends Applet {
   Label userID, password;
   TextField userText, passText;
   Button submitBtn;

   public void init() {
      setLayout(new GridBagLayout());
      GridBagConstraints gbc = new GridBagConstraints();
      gbc.insets = new Insets(10, 10, 10, 10);

      // Username Label and TextField
      userID = new Label("Enter UserName:");
      gbc.gridx = 0;
      gbc.gridy = 0;
      add(userID, gbc);

      userText = new TextField(15);
      gbc.gridx = 1;
      add(userText, gbc);

      // Password Label and TextField
```

```
        password = new Label("Enter Password:");
        gbc.gridx = 0;
        gbc.gridy = 1;
        add(password, gbc);

        passText = new TextField(15);
        passText.setEchoChar('*'); // Mask the password field
        gbc.gridx = 1;
        add(passText, gbc);

        // Submit Button
        submitBtn = new Button("Submit");
        gbc.gridx = 0;
        gbc.gridy = 2;
        gbc.gridwidth = 2;
        add(submitBtn, gbc);

        // Action Listener for the Button
        submitBtn.addActionListener(new ActionListener() {
           public void actionPerformed(ActionEvent e) {
              String name = "GRENA";
              String pass = "GRENA@120";

              if (userText.getText().equals(name) && passText.getText().equals(pass)) {
                 System.out.println("Details successfully submitted... You are logged in.");
              } else {
                 System.out.println("Invalid data... Please enter valid credentials.");
              } }
     });
   }
}
```